



Workload categories. This guide highlights partner brands sold in India – Samsung, Western Digital/SanDisk, Seagate, Crucial/Micron and ADATA XPG – and categorises drives by typical use:

- **Mainstream gaming and productivity:** Flagship PCIe Gen 4 drives like Samsung's 990 PRO and Western Digital's SN850X offer sequential reads around 7.5 GB/s and writes around 6.9 GB/s. They include DRAM caches and come in capacities up to several terabytes. However, they can run hot under sustained writes; ensure adequate airflow.
- **Balanced Gen 5 options:** Drives like Crucial P510 provide Gen 5 speeds while remaining cool and affordable. Samsung 990 EVO Plus uses a hybrid Gen 5 ×2/Gen 4 ×4 interface to offer mid-Gen 5 performance with low heat. These models are ideal if you want some future-proofing but don't need the fastest available.
- **High-end workstations:** If you work with 8K video or large datasets, premium Gen 5 drives such as Crucial T705 and WD SN8100 push sequential reads to roughly 14 GB/s and writes above 12 GB/s. They use cutting-edge TLC NAND and advanced controllers and require good cooling due to ~11 W power draw. Such drives are overkill for everyday gaming but can drastically reduce time spent waiting for file transfers in professional workflows.
- **Budget and large capacity:** Older Gen 4 drives like WD Blue SN5100 deliver respectable speeds at a lower price and are ideal for secondary storage or budget builds. QLC-based SSDs such as those in ADATA's portfolio offer capacious 2–4 TB

options at affordable prices but suit read-heavy workloads. Another solid performer is Seagate's FireCuda 540 (Gen 5), which pairs high endurance with capacities up to 4 TB. It performs close to the mainstream Gen 5 drives and comes with a thick heatsink. ADATA's XPG GAMMIX S70 Blade (Gen 4) targets compact builds; its slim heatsink makes it suitable for laptops and PlayStation 5 consoles, and it delivers around 7.4 GB/s reads. These models highlight that high capacity does not always mean slow speed when properly designed.

When comparing models, pay attention to price per gigabyte, thermal management (does it come with a heatsink?), warranty length, and features like DRAM cache or hardware encryption. The difference between a 7 GB/s Gen 4 drive and a 14 GB/s Gen 5 drive is most noticeable when copying very large files; for games and general productivity, the cheaper, cooler option may feel just as responsive.

Conclusion

Fast storage can transform your PC experience, but chasing the highest sequential numbers isn't always necessary. PCIe Gen 4 drives deliver near-instant loading for most tasks, while Gen 5 drives are valuable in specific professional roles. Understanding NAND types helps balance cost, capacity and longevity, and endurance ratings provide realistic expectations. Good caching and adequate cooling ensure sustained performance. With careful selection, your SSD will not only keep pace with modern workloads but also extend the lifespan and capability of your entire system. 